

TECHNICAL SPECIFICATIONS FOR SWITCHING SERVICES <Apply for Member Organization>

PART V: ANNEX

NATIONAL PAYMENT CORPORATION OF VIETNAM

Ha Noi, December, 2016



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REVISION HISTORY

Version	Release Date	Editor	Reviewer	Description for change
0.9	From 01/2016 - 11/2016	Ha Nam Ninh Nguyen Hung Cuong Bui Thi Kim Dung Dao Thanh Son Huynh Cong Linh Pham Minh Ngoc Le Anh Tuan Nguyen Thanh Quynh	Nguyen Hung Nguyen	- Built draft Technical Specification Document - Spitted the document to 05 parts include: • Part I – Transaction Processing • Part II – Message format • Part III – File format • Part IV – Security and Communication • Part V – Annex - Added international transaction processing on following part: • Part VI – Annex: International Transaction Processing
1.0	12/2016	Ha Nam Ninh Nguyen Hung Cuong Nguyen Thanh Quynh Dao Thanh Son Huynh Cong Linh	Nguyen Hung Nguyen	- Added some data elements on CHIP transaction at: • Part II – Message Format: Section 6 – Data elements, Section 7 – Message format • Part V – Annex: Section 4 – Response code - Updated the Online switching,

				Tokenization process flow synchronized with regulations for member organizations participating in the NAPAS system at: • Part I – Transaction processing, Section 4.2, 4.3, 4.5 • Part II – Message structure, Section 6.2 - Added service and interchange fees between NAPAS and Member Banks; added two fields : DE 22, DE 25 at : • Part III – Message structure : section 4.3
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1 Audience

“Technical Specifications on NAPAS System Connectivity” document is intended for the Member Organizations of NAPAS includes but not limited to Member Banks of NAPAS and Payment Intermediaries that directly connected to card switching services.

2 Scope

The card switching service allows customers to use Member Banks card in NAPAS network to perform transactions on the device / card accepted channels of NAPAS Member Institution as well as International Card Organizations or Switching Organizations of other countries and vice versa.

This document describes the transaction processing of the switching services that NAPAS deploys to Member Institutions.

Switching services include but not limited to:

2.1 Domestic card switching service

- a. Domestic card switching services at ATM: allow customers to use Member Banks card to perform transactions on the ATM equipment of the Member Banks in the Switching Organizations of other countries, and vice versa. Domestic card switching services on the ATM include but not limited to the following transactions:
 - Balance Inquiry
 - Cash Withdrawal
 - InTra-bank Fund Transfer (ITFT)
 - Mini Statement
 - PIN Change
- b. Domestic card switching service at POS: allow customers to use Member Banks card to perform transactions at POS of Member Bank of NAPAS network or Member Bank of Switching Organizations of other countries, and vice versa. Card switching services at POS include but not limited to the following transactions:
 - Purchase
 - Void
 - Balance Inquiry

- PIN Change
- c. Online switching service (Ecom): allow customers to make purchase through the online payment website that is connected to NAPAS. A successful payment transaction consists of 03 messages after sending to Member Bank:
- Verify Cardholder information (Verify Card)
 - Verify OTP
 - Ecom Purchase

Beside normally online payment service, NAPAS also debits on customer account by processing only one message:

- Ecom Fast Purchase
- d. Inter-Bank Fund Transfer 24/7 service (IBFT): allows the customers to transfer money on many different payment channels (ATM, Internet banking, Mobile banking, Bank counters ...). In addition, this is also a service that Payment Intermediaries can use to allow a fund transfer from e-wallet account of Payment Intermediaries to account of Member Bank of NAPAS. This services include following messages:
- Beneficiary cardholder / account information inquiry (IBFT Inquiry)
 - Beneficiary cardholder / account fund transfer (IBFT Deposit)

Note: IBFT Inquiry message and IBFT Deposit message are 02 independent messages and NAPAS system does not check the relationship between these messages.

- e. Payment code service: allows customer to use Member Bank payment channels which transfers and receives using Payment code (PC). Customer who transfers money is provided a Payment code when perform transfer transaction, beneficiary customer use this Payment code on channel of Bank to receive money. On-us transfer or off-us transfer bank and receive bank may be on the same bank or two different banks. The beneficiary customer object of service mainly direct to customers who have not got bank account.
- Payment code create request (PC Create)
 - Payment code authentication (PC Verify)
 - Payment code withdrawal (PC Cash Withdrawal)
 - Payment code inquiry (PC Inquiry)
- f. Tokenization service: the service that encrypt customer information to Token ID range – a range codes to identify alternative using of customer information when conducting payment.

- Card information verify (Token Verify Card)
- OTP verify (Token Verify OTP)
- Token Purchase
- Token Fast Purchase

2.2 Cross-border card switching services

- a. Cross-border card switching services at ATM/POS: allows customer to use the card of NAPAS Member Bank to perform transactions on the Member Banks ATM/POS equipment of the Switching Organization of other countries, and vice versa. Cross-border card switching services at ATM/POS include but not limited to the following transactions:
 - Balance Inquiry at ATM
 - Cash Withdrawal at ATM
 - Balance Inquiry at POS
 - Purchase at POS
- b. Cross-Border Fund Transfer – CBFT: allows customers to use their bank cards in Switching Organizations of other countries can transfer money of account number/client card number of Member bank in NAPAS. CBFT service deploy to Member Bank based on the platform of the IBFT service, include two transactions:
 - Beneficiary cardholder / account information inquiry (CBFT Inquiry)
 - Beneficiary cardholder / account fund transfer (CBFT Deposit)

3 Terms and Acronyms

The following table describes the terminology and acronyms used on this document:

No.	Terms/Acronyms	Description
1	SW	Switching system (Switch) of NAPAS
2	ACQ	Acquirer (Settlement Bank)
3	ISS	Issuer
4	BNB	Beneficiary Bank
5	PMI	Payment Intermediary
6	MO	Member organization (Include Member Bank and Payments Intermediary)
7	Bank MO	Bank Member Organization
8	ICA	International Card Association (include Visa, MasterCard, Amex, JCB, etc. ...)
9	ATM	Automated Teller Machine
10	POS	Point Of Sale/Service
11	ITFT	InTra-bank Fund Transfer (Transfer to internal bank)
12	IBFT	Inter-Bank Fund Transfer 24/7
13	CBFT	Cross-Border Fund Transfer
14	OTP	One Time Password
15	APN	Asian Payment Network
16	MEPS	Malaysia Electronic Payment System
17	KTFC	Korea Financial Telecommunications & Clearings Institute
18	NETS	Network For Electronic Transfer
19	ITMX	Interbank Transaction Management and Exchange

4 Response code

Response code table descriptions:

Response code	Descriptions	Action	Condition
00	Approved Successfully	Approve	
01	Refer to card issuer	Decline	- The transaction is decided by Issuer and requested to refer to Issuer
03	Invalid Merchant	Decline	- MCC is not allowed. - MCC is in blacklist.
04	Pick-up	Pick-up card	- Issuer request to pick-up the card
05	Unable To Process	Decline	- Invalid transaction time. - Invalid cardholder information. - Un-identified error. - Online transaction services is not registered.
12	Invalid transaction	Decline	- No original transaction (in reversal processing) - Invalid original transaction informations
13	Invalid amount	Decline	- Invalid transaction amount. - Exceed limit transaction amount. - Transaction currency not allowed.
14	Invalid Card number	Decline	- Invalid Card number. - The Check-digit is wrong.

			- The card number in request and response message is not match. - The card number in reversal and original message is not match. - Invalid payment code. - Invalid beneficiary card/account
15	No such Issuer	Decline	- Issuer bank is not be found based on request information (card number, account number...)
21	Card not initialized	Decline	- Card is not activated / or invalid. - Created PIN require to be changed. - Card is closed (not use for limited time).
25	Unable to locate original transaction	Decline	- The card number in request and response message is not match.
30	Message Format Error	Decline	- Cannot read / analyze transaction data. - Cannot read/ analyze sub- field data. - Cannot find request field data. - Invalid characters value in field data. - Invalid tracks data.
34	Suspected Fraud	Pick-up card	- Card is identified as fraud, need to be captured.
39	No credit account	Decline	- Credit account is invalid.
41	Lost Card	Pick-up card	- Card is lost by customer. Card capture is required.

43	Stolen Card	Pick-up card	- Card has been stolen, capture card.
51	Insufficient Balance	Decline	- Insufficient fund balance, transaction can't be completed. - Credit account is over limit.
53	No saving account	Decline	- Invalid saving account.
54	Expire Card	Decline	- Expiry card. - Invalid card expiry date. - Payment code has been expired. - OTP code has been expired. - Entered PIN is invalid. - Input wrong OTP.
55	Incorrect PIN	Decline	- Invalid PIN. - Invalid OTP code.
57	Transaction not permitted to cardholder	Decline	- Transaction not permitted to cardholder. - Off-us transaction is not permitted. - Fallback transaction is not allowed by Issuer.
58	Transaction not permitted to terminal	Decline	- Transaction is not permitted to terminal (rejected by issuer). - Terminal ID not match.
59	Suspected Fraud	Decline	- Suspected transaction, transaction should be declined.
61	Exceeds withdrawal amount limit	Decline	- Amount per transaction is over limit.

			- Acceptance card transaction amount is over limit. - Amount transaction per day is over limit.
62	Restricted Card	Decline	- Transaction is declined by restriction policy.
63	Security Violation	Decline	- MAC process error. - PIN translate error. - PIN process error. - Invalid Key length. - Invalid PIN length. - Declided by Security Policy. - CVV process error.
64	Original Amount Incorrect	Decline	- Transaction amount of request and response messages not match. - Transaction amount of original and reversal transaction not match.
65	Exceed withdrawal frequency limit	Decline	- Over limit of withdrawal per day. - Cash withdrawal or purchase per day limit is exceed
68	Response received too late (time-out)	Decline	- Timeout by Issuer.
75	Allowable number of PIN tries exceeded	Decline	- PIN tries limit is exceed - Wrong input OTP over 03 times. Payment code is locked. - Wrong input OTP over 03 times. Transaction is decided.
76	Invalid Account	Decline	- Invalid Account.

84	ARQC validation error	Decline	- ARQC validation error.
85	No CVM Threshold exceeded, enter PIN	Decline	- No CVM is declined by Issuer, PIN enter is required.
90	Cut-off is in progress	Decline	- Issuer system is in cut-off processing. - Beneficial Bank system is in Cut-off processing.
91	Issuer or switch is in operation	Decline	- Issuer is not ready. - Sign-off is sent from Issuer to Napas. - Napas and Issuer host is not establish. - Beneficial bank is not ready.
92	Financial institution or intermediate network facility cannot be found for routing	Decline	- Network to issuer is not available. - Network to beneficiary bank is not available.
94	Duplicate transaction	Decline	- Original transaction is Duplicate, declined transaction.
96	System malfunction	Decline	- Transaction is declined by Napas. - Napas system is under maintenance.

Other case, to ensure compatibility with NAPAS and member system, in the first phase of upgraded technical standards, NAPAS still support some Response code as bellow:

Response code	Description	Action
02	Refer to Issuer Bank	Decline
04	Do not Honor	Pick-up

06	Invalid transaction terminal	Decline
08	Issuer Time out	Decline
09	No Original	Decline
10	Unable to Reverse	Decline
17	Invalid capture date	Decline
19	System error re-enter	Decline
20	No From Account	Decline
21	No To Account	Decline
22	No Checking Account	Decline
23	No Saving Account	Decline
24	No Credit Account	Decline
25	Unable to locate record	Decline
39	Transaction Not Allowed	Decline
42	Special Pickup	Decline
66	Exceeds Acquirer Limit	Decline
79	Key validation Error	Decline
80	Pin Length Error	Decline
81	Invalid Pin Block	Decline
82	Invalid CVV	Decline
83	Counter Sync Error	Decline
87	PIN Key Error	Decline
88	MAC Sync Error	Decline
93	Invalid Acquirer	Decline
94	Duplicate Transaction IBFT	Decline
98	Duplicate Reversal	Decline
99	Duplicate Transaction	Decline

Table of Response code based on services using NAPAS system:

Res pons e Cod e	Description	ATM/POS services		IBFT servi ces	Onlin e switc hing and Token izatio n servic es	Payment Code services	
		0210	0430	0210	0210	0210	0430
00	Approved Successfully	X	X	X	X	X	X
01	Refer to card issuer	X			X		
03	Invalid Merchant	X		X			
04	Pick-up	X					
05	Unable To Process	X		X	X	X	
12	Invalid transaction	X		X	X	X	
13	Invalid amount	X		X	X	X	
14	Invalid Card number	X		X	X	X	
15	No such Issuer	X					
21	Card not initalized	X					
25	Unable to locate original transaction	X		X			
30	Message Format Error	X		X	X	X	
34	Suspected Fraud	X					
39	No credit account	X		X			
41	Lost Card	X			X		

43	Stolen Card	X			X		
51	Insufficient Balance	X			X		
53	No saving account			X			
54	Expire Card	X		X	X	X	
55	Incorrect PIN	X			X	X	
57	Transaction not permitted to cardholder	X					
58	Transaction not permitted to terminal	X					
59	Suspected Fraud	X		X			
61	Exceeds withdrawal amount limit	X		X	X		
62	Restricted Card	X					
63	Security Violation	X		X	X		
64	Original Amount Incorrect	X		X			
65	Exceed withdrawal frequency limit	X		X	X		
68	Response received too late (time-out)	X		X	X		
75	Allowable number of PIN tries exceeded	X			X	X	
76	Invalid Account	X		X	X		
90	Cut-off is in progress	X		X			

91	Issuer or switch is in operation	X		X	X	X	
92	Financial institution or intermediate network facility cannot be found for routing	X		X			
94	Duplicate transaction	X	X	X	X	X	X
96	System malfunction	X		X	X		

5 Regulation on Merchant Category Code (MCC)

Table of Merchant Category Code for the Member Bank who implement NAPAS switching service:

MCC	Group	Description	MCC categories corresponding to international standard
0700	Agriculture	Veterinary medicine ; Spirit Producers ; Wine Producers ; Agriculture; ...	0001-1499
4000	Transportation	Railroads ; Taxicabs and Limousines ; Bus Lines ; Boat Rentals and Leasing ; Tolls and Bridge Fees ; Airports, Flying Fields, and Airport Terminals. ...	4000-4799
4111		Transportation	
4112		Passenger Railways	
4131		Bus Lines	
4789		Transportation Services - Not included Railways, Airport.	
4784		Tolls and Bridge Fees	
4800	Utilities Services	Telecommunications Services ; Cable and Other Pay Television Services ; Telegraph Services ; Computer Network Services ; Utilities – Electric ;	4800-4999

		Water ; Gas ; Cablegram ;	
4814		Telecommunications Services ;	
4816		Computer Network / Information Services	
4899		Cable and Other Pay Television Services ;	
4900		Utilities – Electric ; Gas ; Water ; Sanitary services ;	
5000	Retail Services	Books, Periodicals, and Newspapers ; Cigar Stores and Stands ; Stationery; Computer Equipment ; Building Materials Stores ; Office and Commercial Furniture; Housing equipment ; Food and Drink ; Florists' Supplies, Nursery Stock, and Flowers ; Mobile store ; Duty free store ; Cake store ; ...	5000-5499
5411		Grocery Stores and Supermarkets	
5500	Automobiles and Vehicles.	Automobiles and Vehicles. Car and Truck Dealers (New and Used) - Sales, Service, Repairs, Parts, and Leasing ; Automotive Tire Stores ; Auto Parts and Accessories Stores ;	5500-5599

		Services station ; Petro station ; Boat Dealers ; Motors Home Dealers ; ...	
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5600	Clothing Stores	Men's and Boy's Clothing and Accessory Stores ; Women's Accessory and Specialty Stores ; Children's and Infant's Wear Stores ; Sports and Riding Apparel Stores ; Family Clothing Stores ; Labour Protection Stores ; ...	5600-5699
5700	Miscellaneous Stores	Furniture Stores; Electronics Stores ; Computer Software Stores ; Music Record Stores ; Fireplaces Stores ; Restaurants ; Bars ; Fast Food Restaurants ; Bicycle Shops ; Watches ; Swimming Pools ; Stamp and Coin Stores ; Pet Shops ; ...	5700-5999
5814		Fast Food Restaurants ;	
6011	Financial Institutions - Automated Cash Disbursements	ATM	
6051	Service Providers	Non-Financial Institutions - Foreign Currency, Money Orders, Scrip, and Travelers' Checks (not Wire Transfers) ; Securities-Brokers and Dealers	6051-7299

		Insurance ; Lodging Hotels, Motels, and Resorts ; Laundry, Cleaning and Garment Services ; Beauty and Barber Shops ; Private hospitals ; Child Care Services ; Clothing Rental - Costumes, Uniforms, and Formal Wear ; Heath and Beauty Spas ; ...	
7300	Business Services	Advertising Services ; Commercial Photography, Art, and Graphics ; Design Services ; Maintenance and Repair Services ; Hire and rent ; ...	7300-7529
7399	Business Services	Commercial Services	
7530	Repair Services	Automotive Body Repair Shop ; Car Washes ; Electronics Repair Shops ; Air Conditioning and Refrigeration Repair Shops ; Watch, Clock Repair; Jewelry Repair Shops ; Furniture Repair ;	7530-7799
7800	Amusement and Entertainment	Bowling Alleys ; Billiard ; Tourist ; Golf ;	7800-7999

		Motion Picture Theaters ; Football Club ; Game services ; Amusement Parks ; Dance Halls, Studios, and Schools ; ...	
8000	Professional Services and membership.	Legal Services and Attorneys ; Architectural, Engineering, and Surveying Services ; Accounting, Auditing, and Bookkeeping Services ; Child Care Services ; Colleges, Universities, Professional Schools, and Junior Colleges ; Dentists and Orthodontists ; Charitable Organizations; Political and Social Service Organizations ; Association ; Hospitals ; ...	8000-8999
8211		Elementary and Secondary Schools	
8220		Colleges, Universities, Professional Schools, and Junior Colleges	
9200	Government Services	Tax Payments ; Fines ; Court Costs ; Postal Services ...	9000-9999
9000	Government Services	Government Services	

6 UPI CHIP Acceptance

6.1 UPI CHIP Acceptance Test Process

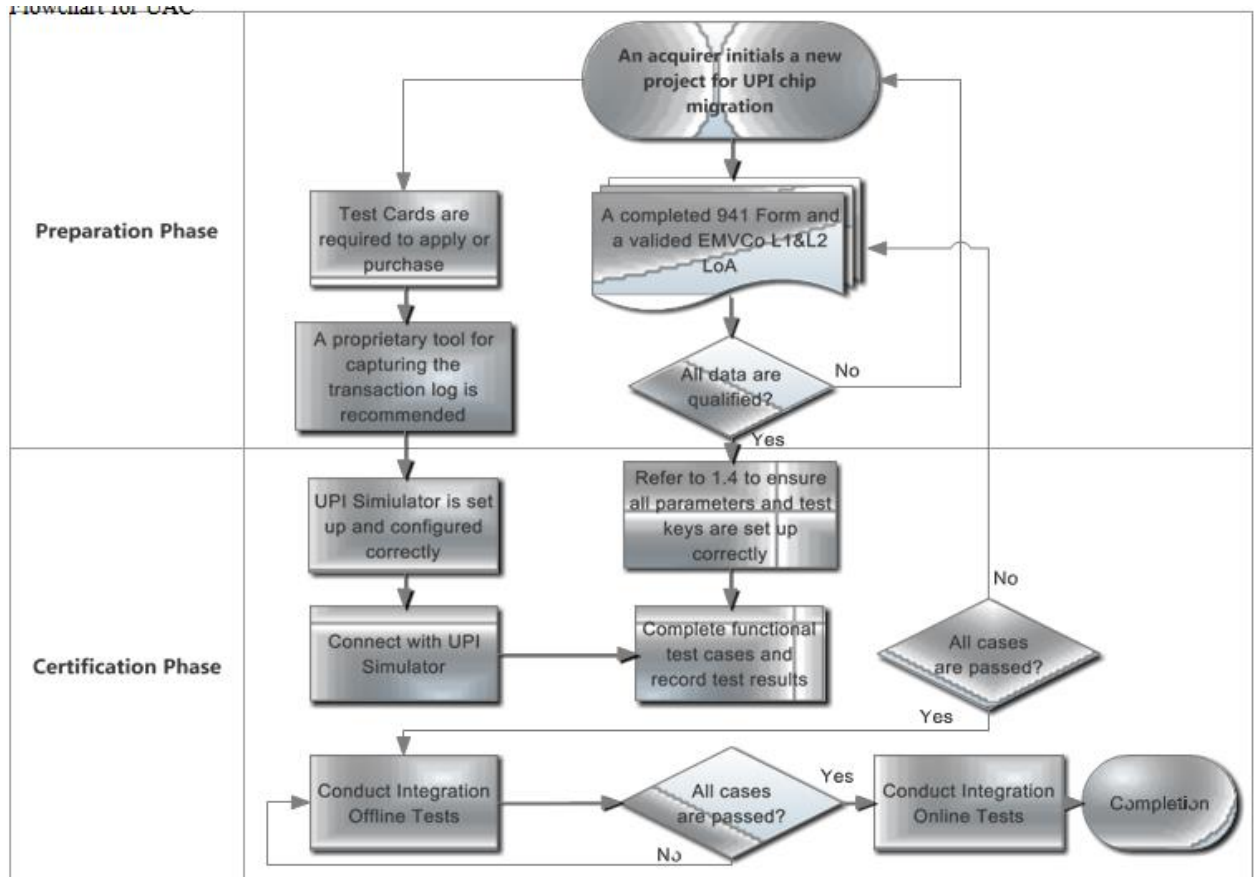


Figure 1. UPI CHIP Acceptance Test Process

6.1.1 Preparation Step

- On this step, Member bank prepares EMV Co L1&L2 certification for each terminal model that accepts UPI CHIP cards. In case Member bank wants to implement how many terminal models, so that they should prepare corresponding EMV Co L1&L2 certifications and fulfills to form 941 that NAPAS send to. About ACQ information, NAPAS will support Member Bank to complete and send to UPI.
- And Member Banks also researchs, learns about test cases for implementing according to NAPAS attachment documents.

6.1.2 Get Certification Step

- On getting certification step, Member Bank will follow NAPAS on 03 testing phases.

- ✓ Function test
- ✓ Integration test
- ✓ Online test

6.1.3 Function Test

- This step assures Credit/Debit application on interface terminal works with all UPI Chip card functions.
- This step will performance testing with simulator provided by UPI at NAPAS.
- The result of this step is transaction statements and test logs from UPI simulator

6.1.4 Integration Test

- This step purpose testing successful integration for transaction processing at terminal by host of NAPAS.
- This step performance with UPI simulator
- The result of this step is test logs from UPI simulator

6.1.5 Online Test

- This step performs testing full operation functions of each chip transaction types.
- Transaction is processed at terminal, thought NAPAS host and sent to ISS of UPI to receive an approval code.
- The result of this step is log tests between NAPAS and UPI
- The preparation steps should be done as follows:

	Works List	Parties
Terminal Reviewing		
	Prepare EMV Certification for each terminal	Bank & Terminal Vendor
	Certification form from UPI	Bank & Terminal Vendor
	Load UnionPay IC application to each terminal	Bank & Terminal Vendor
	Load Root CA key and other parameter.	Bank & Terminal Vendor
Testing agreement		
	Require License for Simulator	NAPAS

	Setup Simulator	Bank & NAPAS
	Analyse test plans	Bank & Host vendor & NAPAS
Terminal testing		
	Perform function test cases	Bank
	Send test results from Simulator to UPI	NAPAS
	Send scanned receipts to UPI	Bank & NAPAS
	UPI authorizes the result by email	NAPAS
Offline testing		
	Perform offline test cases by simulator	Bank
	Send offline test result from Simulator to UPI	NAPAS
	UPI authorizes the result by email	NAPAS
Online testing		
	Register online test plan with UPI	NAPAS
	Perform test cases	Bank & NAPAS
	Do exception test cases, need support from UPI	Bank & NAPAS
	Settlement and Reconciliation	Bank & NAPAS
	Verify test result and settlement	Bank & NAPAS
	UPI provide certification	Bank & NAPAS
Launching		
	Send launching request	Bank & NAPAS
	Golive on production environment	Bank & NAPAS

6.2 ARQC/ARPC Processing

- At this time, UPI does not support Verify PIN Offline
- And UPI has 02 chip card profiles as E-Cash and Debit/Credit Application.
- ARQC: script is sent from terminal to ISS after performing communication with chip card

- ARPC: script is sent from ISS responds to terminal after receiving the request



Figure 2. ARQC/ARPC processing flow

- The decision of transaction is gone online or offline is due to UPI CHIP application.
- After communicate with card, a terminal needs to verify transaction online that the terminal will generate ARQC code and send to ISS though intermediate host.
- ISS processes ARQC code and responds an ARPC code contains following actions:
 - ✓ Application Block/Unblock
 - ✓ Card Block/Unblock
 - ✓ PIN Change / Unblock
 - ✓ Update Data
 - ✓ Update Record
- Terminal will use an ARPC to update application on UPI card.
- After that, terminal will initiate a 0620/0630 message to inform the script running result to ISS. But now, NAPAS system does not process this transaction type.

7 Specification on BIN assignment of Bank

Each member bank that implement services with NAPAS should have an assigned BIN (Bank Identification Number).

Assigned BIN of BANKs:

- Card BIN numbering system complies with ISO 7812 and ISO 3166

Structure of Card number

- When being issued, each card has a defined number (or “card number”). The structure of “Card number” is as follows:

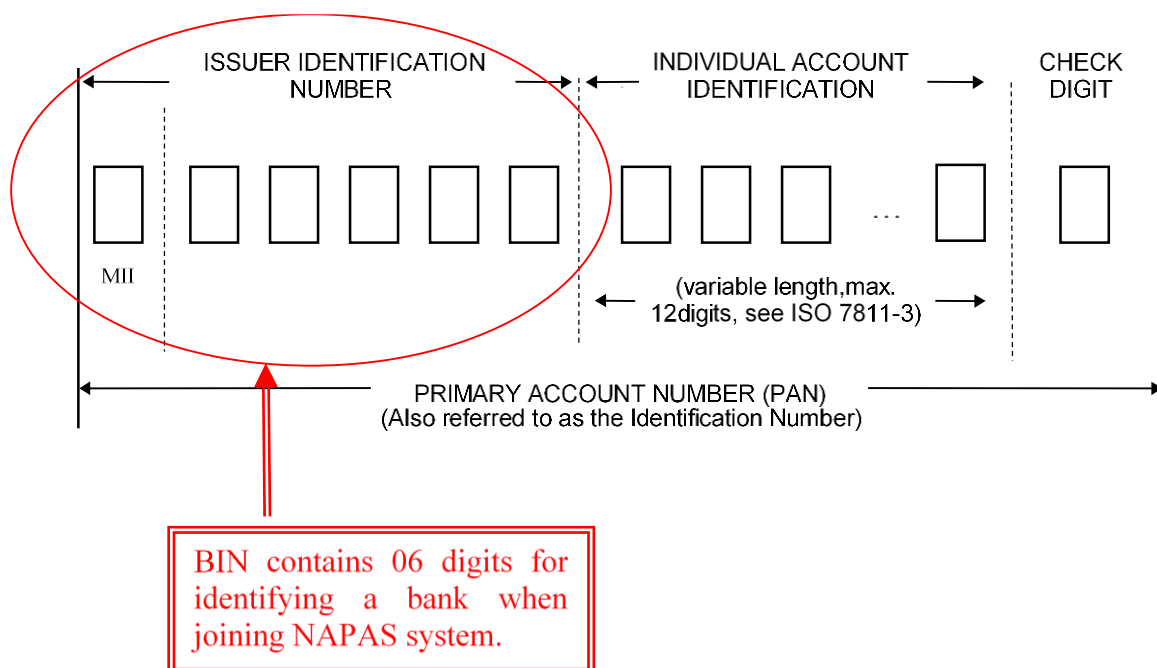


Figure 3. Card number structure

MI (Major Industry Identifier) – is the identification number of industrial services.

When joining in NAPAS system, the digits after BIN are defined by each Member Bank.

BIN (Bank Identification Number), or IIN (Issuer Identification Number) for card issuing institutions, has format as follows:

9	7	0	4	x	x
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- The first digit is **9** to indicate local cards.
- The next 03 digits are **704** to indicate country code of Vietnam.
- The last 2 digits (**xx**) have optional value defined and assigned for each Member.
- When joins NAPAS switching system, Member Banks can keep their former BIN to avoid changing cards for customers. However, for new cards to be issued, banks should follow the regulations on BIN numbering as indicated in the Table below.
- Banks agree to register BIN with the format 9704**xx**, where **xx** are the digits selected and registered by banks

Following list of Bank BIN that implements NAPAS's services:

No.	Trading name	Bank's name	Switching		PAN Length
			ACQ ID	BIN LIST	
1	Bao Viet Bank	Bao Viet Commercial Joint Stock Bank	970438	970438	16
2	Eximbank	Vietnam Export – Import Commercial Joint Stock Bank	452999	970431	16
3	Hongleong Bank	Hong Leong Bank Vietnam	970442	970442	16
4	Indovina Bank	Indovina Bank Ltd.	888999	970434	16
5	Kien Long Bank	Kien Long Commercial Joint Stock Bank	970452	970452	16
6	Lien Viet Post Bank	LienViet Post Commercial Joint Stock Bank	970449	970449	19
7	MBBank	Military Commercial Joint Stock Bank	970422	970422	16
8	Maritime Bank	Maritime Commercial Joint Stock Bank	970426	970426	16
9	NASB	North Asia Commercial Joint Stock Bank	970409	970409	16
10	NCB	National Citizen Bank	818188	970419	16
11	OCB	Orient Commercial Joint Stock Bank	970448	970448	16
12	SCB	Sai Gon Joint Stock Commercial Bank	157979	970429	16
13	SHB	Saigon-Hanoi Commercial Joint Stock Bank	970443	970443	16
14	Standard Chartered Vietnam	Standard Charter Bank Vietnam	970410	970410	16
15	Shinhan Vietnam	Shinhan Vietnam Bank	970424	970424	16

16	Techcombank	Viet Nam Technological and Commercial Joint Stock Bank	888899	889988	16
				970407	16
17	TPBank	Tien Phong Bank	970423	970423	16
18	Viet A Bank	Viet A Joint Stock Commercial Bank	166888	970427	16
19	Vietcombank	Joint Stock Commercial Bank for Foreign Trade of Vietnam	686868	686868	16
				970436	19
20	Viet Capital Bank	Viet Capital Commercial Joint Stock Bank	970454	970454	16
21	VIBank	Vietnam International Commercial Joint Stock Bank	180906	180906	16
				970441	19
22	VID Public Bank	VID Public Bank	970439	970439	16
23	VPBank	Vietnam Prosperity Commercial Joint Stock Bank	981957	970432	16
24	ABBANK	An Binh Commercial Joint Stock Bank	970459	970425	16
25	ACB	Asia Commercial Bank	970416	970416	16
26	GP.Bank	Global Petro Commercial Joint Stock Bank	970408	970408	16
27	HDBank	Ho Chi Minh City Development Joint Stock Commercial Bank	970437	970437	16
28	SeABank	Southeast Asia Commercial Joint Stock Bank	970468	970440	19
29	Sacombank	Sai Gon Thuong Tin Commercial Joint-stock Bank	970403	970403	16

30	Agribank	Vietnam Bank for Agriculture and Rural Development	970499	970405	16
31	BIDV	Bank for Investment and Development of Vietnam	970488	970418	16
32	Vietinbank	Vietnam Bank for Industry and Trade	970489	970415	16
				62016	16
33	Saigonbank	Saigon Bank for Industrial and Trade	161087	970400	16
34	OceanBank	OCEAN Commercial Joint Stock Bank	970414	970414	16 & 19
35	VRB	Vietnam – Russia Joint Venture Bank	970421	970421	16
36	PVcomBank	Viet Nam Public Bank	970412	970412	16
37	PG Bank	Petrolimex Group Commercial Joint Stock Bank	970430	970430	16
38	CBBank	Vietnam Construction Bank	970444	970444	19
39	Nam A Bank	Nam A Commercial Joint Stock Bank	970428	970428	16
40	Co-opBank	Cooperative Bank of Vietnam	970446	970446	19
41	Dong A Bank	DongA Joint Stock Commercial Bank	970406	970406	16
42	IBK HN	Industrial Bank of Korea - Hanoi Branch	970455	970455	16
43	IBK HCM	Industrial Bank of Korea - HCM Branch	970456	970456	16
44	UOB	United Overseas Bank	455372	UOB implements only POS Acquiring	

8 Specification on Magnetic Card and Card Reading Terminal

8.1 Physical characteristics of cards

Physical characteristics of magnetic cards (size, weight, material structure, hardness, plasticity, strength...) comply with ISO 7810.

8.2 Embossed information on cards

Embossed information on cards complies with ISO 7811-1

Information recorded on magnetic stripe of cards and structure of information fields comply with ISO 7811 and ISO 7813.

8.3 Track 1 Data Format

Code format forming characters on Track 1 is defined in ISO 7811-2 or ISO 7811-6 (Appendix 3 - attached). Each character is formed by 7 bits including 6 data bits + 1 bit for odd-even checking. Maximum number of characters is 64, including 36 alphanumeric characters, 3 characters for defined fields (Start Sentinel, End Sentinel and Field Separator), the last 25 characters are special characters. The following table describes the 64 characters in details:

Data bits						B 7	Character	Value (Hex)	Function
b 1	b 2	b 3	b 4	b 5	b 6				
0	0	0	0	0	0	1	Space	00	Special
1	0	0	0	0	0	0	!	01	Special
0	1	0	0	0	0	0	"	02	Special
1	1	0	0	0	0	1	#	03	Special
0	0	1	0	0	0	0	\$	04	Special
1	0	1	0	0	0	1	%	05	Start Sentinel
0	1	1	0	0	0	1	&	06	Special
1	1	1	0	0	0	0	'	07	Special
0	0	0	1	0	0	0	(	08	Special
1	0	0	1	0	0	1	)	09	Special
0	1	0	1	0	0	1	*	0A	Special
1	1	0	1	0	0	0	+	0B	Special

Data bits						B 7	Character	Value (Hex)	Function
b 1	b 2	b 3	b 4	b 5	b 6				
0	0	1	1	0	0	1	,	0C	Special
1	0	1	1	0	0	0	-	0D	Special
0	1	1	1	0	0	0	.	0E	Special
1	0	0	1	0	0	1	/	0F	Special
0	0	0	0	1	0	0	0	10	Data
1	0	0	0	1	0	1	1	11	Data
0	1	0	0	1	0	1	2	12	Data
1	1	0	0	1	0	0	3	13	Data
0	0	1	0	1	0	1	4	14	Data
1	0	1	0	1	0	0	5	15	Data
0	1	1	0	1	0	0	6	16	Data
1	1	1	0	1	0	1	7	17	Data
0	0	0	1	1	0	1	8	18	Data
1	0	0	1	1	0	0	9	19	Data
0	1	0	1	1	0	0	:	1A	Special
1	1	0	1	1	0	1	;	1B	Special
0	0	1	1	1	0	0	<	1C	Special
1	0	1	1	1	0	1	=	1D	Special
0	1	1	1	1	0	1	>	1E	Special
1	1	1	1	1	0	0	?	1F	End sentinel
0	0	0	0	0	1	0	@	20	Special
1	0	0	0	0	1	1	A	21	Data
0	1	0	0	0	1	1	B	22	Data
1	1	0	0	0	1	0	C	23	Data
0	0	1	0	0	1	1	D	24	Data
1	0	1	0	0	1	0	E	25	Data
0	1	1	0	0	1	0	F	26	Data
1	1	1	0	0	1	1	G	27	Data
0	0	0	1	0	1	1	H	28	Data

Data bits						B 7	Character	Value (Hex)	Function
b 1	b 2	b 3	b 4	b 5	b 6				
1	0	0	1	0	1	0	I	29	Data
0	1	0	1	0	1	0	J	2A	Data
1	1	0	1	0	1	1	K	2B	Data
0	0	1	1	0	1	0	L	2C	Data
1	0	1	1	0	1	1	M	2D	Data
0	1	1	1	0	1	1	N	2E	Data
1	1	1	1	0	1	0	O	2F	Data
0	0	0	0	1	1	1	P	30	Data
1	0	0	0	1	1	0	Q	31	Data
0	1	0	0	1	1	0	R	32	Data
1	1	0	0	1	1	1	S	33	Data
0	0	1	0	1	1	0	T	34	Data
1	0	1	0	1	1	1	U	35	Data
0	1	1	0	1	1	1	V	36	Data
1	1	1	0	1	1	0	W	37	Data
0	0	0	1	1	1	0	X	38	Data
1	0	0	1	1	1	1	Y	39	Data
0	1	0	1	1	1	1	Z	3A	Data
1	1	0	1	1	1	0	[	3B	Special
0	0	1	1	1	1	1	\	3C	Special
1	0	1	1	1	1	0	]	3D	Special
0	1	1	1	1	1	0	^	3E	Field Separator
1	1	1	1	1	1	1	_	3F	Special

8.4 Track 2 Data Format

Code format forming characters on Track 2 is defined in ISO 7811-2 and ISO 7811-6. Each character on Track 2 is formed by 5 bits, containing 4 bits for data + 1 bit for odd-even checking. Maximum number of characters is 16. In these 16 characters, there are 10 numeric characters, 3 characters for identification; the

last 3 characters are special characters. The following table describes the 16 characters in details:

Data bits				B 5	Character	Value (Hex)	Function
B 1	b 2	b 3	b 4				
0	0	0	0	1	0	00	Data
1	0	0	0	0	1	01	Data
0	1	0	0	0	2	02	Data
1	1	0	0	1	3	03	Data
0	0	1	0	0	4	04	Data
1	0	1	0	1	5	05	Data
0	1	1	0	1	6	06	Data
1	1	1	0	0	7	07	Data
0	0	0	1	0	8	08	Data
1	0	0	1	1	9	09	Data
0	1	0	1	1	:	0A	Control
1	1	0	1	0	;	0B	Start Sentinel
0	0	1	1	1	<	0C	Control
1	0	1	1	0	=	0D	Field Separator
0	1	1	1	0	>	0E	Control
1	1	1	1	1	?	0F	End Sentinel

8.5 Data structure on tracks

Information is recorded on magnetic cards as follows:

- Information on magnetic cards shall be recorded on Track 1 and Track 2 (or Track 2);
- Structure of data fields on Track 1 and Track 2 complies with ISO 7813 (Appendix 5 - attached), is basically described as follows:

Track 1:

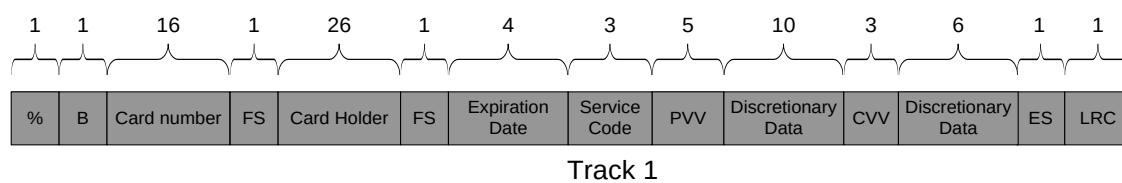


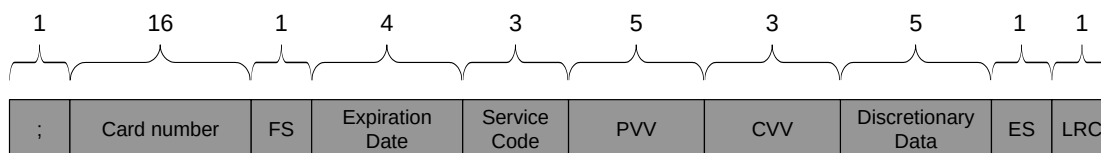
Figure 4. Track 1 data structure

No.	Field	Length	Value	Description
1	Start Sentinel (SS)	1	%	Contain the start sentinel of Track 1
2	Format Code (FC)	1	B	Contain character defining card using areas. For financial and banking area, this character is defined as "B"
3	Card Number	16		Indicate card number of customer. Card number structure is defined by each bank. This field contains various sub-fields such as: BIN, branch code ...
4	Field Separator (FS)	1	^	Separate information fields.
5	Card Holder	26		Contain information of card holder's name.
6	Field Separator (FS)	1	^	Separate information fields.
7	Expiration Date	4		Define card expiration date. Format of this field is YYMM (year, month).
8	Service Code	3	xxx	Define different service categories. The first character defines interchange areas as national/international or only within the bank (intrabank); the second character defines authorization process, available for online authorization by card issuing institution or by the third party with the approval of card issuing institution; the third character defines service range and PIN requirements.
9	Offset number or PIN Verification Value (PVV)	5	"1"+xxxx "0"+0000	This field may or may not have values – as defined by the first character of the field.

10	Discretionary Data	10	*****	Containing private information of each bank which is defined by card issuing bank.
11	Card Verification Value (CVV)	3	xxx	Cryptographic value created from card number, expiration date and service code.
12	Discretionary Data	6	*****	Containing private information of each bank which is defined by card issuing bank.
13	End Sentinel (ES)	1	?	Character marking end of Track 1
14	LongituBM #Redundancy Check Disk (LRC)	1	*	Character created for checking data recorded on Track 1.

(*) Characters may have arbitrary value.

Track 2:



Track 2

Figure 5. Track 2 data structure

No.	Field	Length	Value	Description
1	Start Sentinel (SS)	1	;	Containing the start sentinel of Track 2
2	Card Number	16		Containing account number of customer account number structure is defined by each bank. This field contains various sub-fields such as: BIN,

No.	Field	Length	Value	Description
				card number, branch code ...
3	Field Separator (FS)	1	=	Separate information fields.
4	Expiration Date	4		Indicate card expiration date. Format of this field is YYMM (year, month).
5	Service Code	3	xxx	Define different service categories. The first character defines interchange areas as national/international or only within the bank; the second defines authorization process, available for online authorization by the card issuing institution or by the third party with the approval of card issuing institution; the third defines service range and PIN requirements.
6	Offset or PIN Verification Value (PVV)	5	"1"+xxxx "0"+0000	This field may or may not have values as defined by the first character. If the first character value is 1, PVV is present, if the first character value is 0, PVV is absent.
7	Card Verification Value (CVV)	3	xxx	Indicate code value created from card number, expiration date and service code.
8	Discretionary Data	5	*****	Field containing private data of each bank which is defined by card issuing bank.
9	End Sentinel (ES)	1	?	Containing the character marking end of Track 2.
10	LongituBM #Redundancy Check Disk (LRC)	1	*	Contain the character created for checking data recorded on Track 2.

Note:

- xxx: means this field is the same on different tracks of cards
- ***: means data will be different on different tracks.

Sample of magnetic cards

Example 1:

- Information embossed on card:

Card holder: TRAN TRI MANH
Card number: 1234 5678 9012 3445
Expiration : 11/06

- Information recorded on Track 1:

```
%B1234567890123445^TRI                                MANH/TRAN
^06111211XXXX*****XXX*****?*
^^^                ^^                ^^    ^    ^^    ^
^                ^^
||_ Card number  ||_ Card holder                ||    |    ||    |_
Discr. |_ CVV    ||_ LRC
||_ Format code  |_ Field separator                ||    |    ||    data
|_ End sentinel
|_ Start sentinel                Field separator _||    |    ||_ PVV*
                                Expiration Date _|    |    |_ PVV key
indicator
                                |_    Service
code
```

- Information recorded on Track 2:

```
;1234567890123445=06111211XXXXXXXXX*****?*
^^                ^^    ^    ^^    ^                ^^
||_ Card number  ||    |    ||    |_ CVV  ||_ LRC
|_ Start sentinel||    |    ||_ PVV*    |_ End sentinel
Field separator _||    |    |_ PVV key indicator
Expiration Date _|    |_ Service code
```

Example 2:

- Information embossed on card:

Card holder: TRAN TRI MANH
Card number: 1234 5678 9012 3445
Expiration : 11/06

- Information recorded on Track 1:

```
%B1234567890123445^TRI                                MANH/TRAN
^06111200000000000000000000000000**XXX*****?*
^^^                ^^                ^^      ^      ^
^                ^^                ^^                ^

||_ Card number    ||_ Card holder                ||    |    |
|_ CVV** ||_ LRC

||_ Format code    |_ Field separator                ||    |    |
|_ End sentinel

|_ Start sentinel                Field separator _||    |    |_
Discretionary data

                                Expiration _|    |    |_ Service
code
```

- Information recorded on Track 2:

```
;1234567890123445=06111201XXXX000000000?*
^^                ^^      ^      ^                ^^

||_ Card number    ||    |    |_ Encrypted||_ LRC
|_ Start sentinel||    |    PIN***    |_ End sentinel
                ||    |_ Service code

Field separator _||_ Expiration
```

Note:

- Service Code in Example 1 is 121, this means the cards are used for international transactions, the transactions are authorized online by card issuing institution, and the cards have no limit to service range;
- PVV Key Indicator: This field indicates whether PVV or Offset value is present;
- Information fields with XXX and of the same color have the same value in each example;
- Information on Track 2 will be reproduced in messages for card transactions going through NAPAS Switch;
- Information on Track 1 is optional.

9 Card Acceptor Terminal

Requirements for ATM/POS which join to NAPAS payment network are as follows:

9.1 Requirements for ATM

- ATM card acceptor terminals are able to read information on both tracks: Track 1 and Track 2;
- Allows to enter PIN as 06 character in maximum
- ATM card acceptor terminal which allows entering PIN has module EPP (Encrypting PIN Pad – the equipment used to encrypt PIN right under keyboard of the ATM) and support 3DES. The encrypted terminal complies with ISO 13491-1 and ISO 13491-2.
- ATM has ability to identify whether BIN recorded on cards belongs to the Issuer joining NAPAS share switching network or not.
- A transaction can be processed successfully at ATMs only when it is authorized by the Issuer.

9.2 Requirements for POS

- POS has ability to choose type of transaction, such as: **Purchase/Sale, Balance Inquiry, Void, and PIN change.**
- POS card acceptor terminals are able to read information on both tracks: Track 1 and Track 2.
- Allows to enter PIN as 06 character in maximum
- POS card acceptor terminal which allows entering PIN has module EPP (Encrypting PIN Pad – the equipment used to encrypt PIN right under keyboard of the ATM) and support 3DES. The encrypted terminal complies with ISO 13491-1 and ISO 13491-2.
- POS terminals allow cards of other member banks to be swiped or manually entered and authorization request must be sent to the Issuer. A transaction can be performed successfully at POS only when it is authorized by the Issuer.
- POS terminals must be configured to contain spaces for cards of Issuers joining the shared switching system so that cards of other member banks are allowed to perform transactions at the terminals.

10 ISO 7810

This International Standard is one of a series of standards describing the characteristics for identification cards as defined in the definitions clause and the use of such cards for international interchange. This International Standard specifies the physical characteristics of identification cards including card materials, construction, characteristics, and dimensions for four sizes of cards.

- The card consists of four discrepancy types of model with 0.76 mm thick (see the following figures for more details)
 - ID-000: 25 mm x 15 mm
 - ID-01: 85.60 mm x 53.98 mm
 - ID-02: 105 mm x 74 mm
 - ID-03: 125 mm x 88 mm

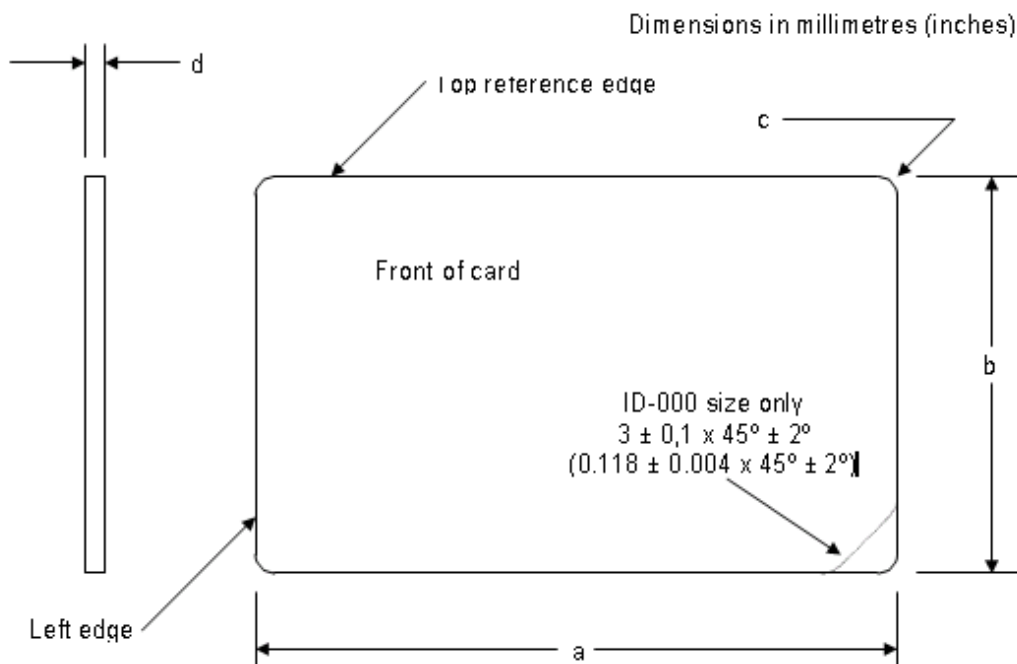


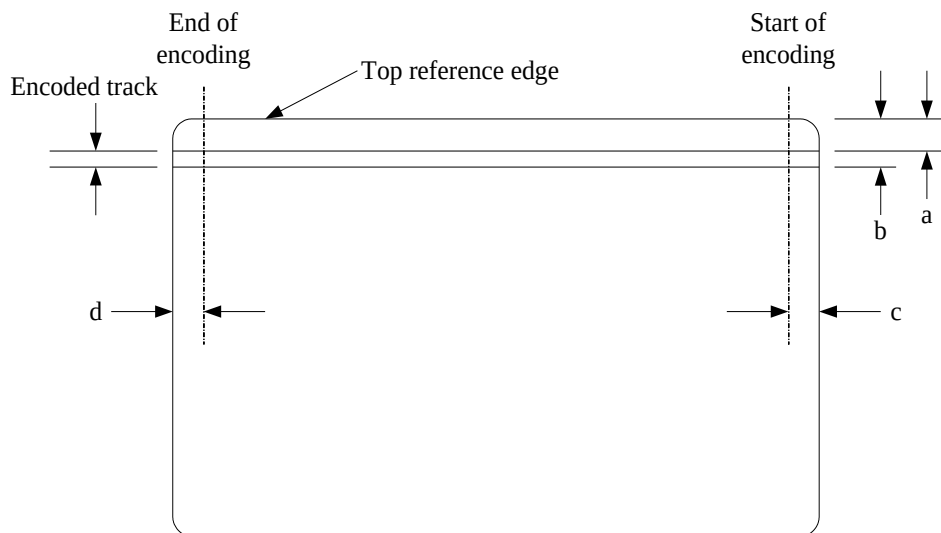
Figure 6. Physical front-site of card structure

	a		b		c		d	
	maximum	minimum	maximum	minimum	maximum	minimum	maximum	minimum
ID-000 Unused card	25,10 (0.988)	24,90 (0.980)	15,10 (0.594)	14,90 (0.587)	1,1 (0.043)	0,9 (0.035)	0,84 (0.033)	0,68 (0.027)
ID-1 Unused card	85,72 (3.375)	85,47 (3.365)	54,03 (2.127)	53,92 (2.123)	3,48 (0.137)	2,88 (0.113)	0,84 (0.033)	0,68 (0.027)
ID-1 Returned card	85,90 (3.382)	85,47 (3.365)	54,18 (2.133)	53,92 (2.123)	3,48 (0.137)	2,88 (0.113)	0,84 (0.033)	0,68 (0.027)
ID-2 Unused card	105,2 (4.142)	104,8 (4.126)	74,2 (2.921)	73,8 (2.906)	5 (0.197)	3 (0.118)	0,84 (0.033)	0,68 (0.027)
ID-2 Returned card	105,3 (4.146)	104,8 (4.126)	74,3 (2.925)	73,7 (2.902)	5 (0.197)	3 (0.118)	0,84 (0.033)	0,68 (0.027)
ID-3 Unused card	125,2 (4.929)	124,8 (4.913)	88,2 (3.472)	87,8 (3.457)	5 (0.197)	3 (0.118)	0,84 (0.033)	0,68 (0.027)
ID-3 Returned card	125,3 (4.933)	124,8 (4.913)	88,3 (3.476)	87,7 (3.453)	5 (0.197)	3 (0.118)	0,84 (0.033)	0,68 (0.027)

Figure 7. Standard card length comply with ISO 7810

- Bedding stiffness: describes on ISO 10373-1 document with maximum bedding stiffness is 35 mm.
- Bearing force: 0.35N/mm in minimum (comply with ISO 10373-1)
- Surface distortion: Raised areas shall not increase the overall card thickness by more than 0,10 mm (0.004 in)
- Card materials: The card shall be made of any material fulfilling the requirements of this International Standard. Card insert material may be used.
- Overall card plane: The maximum distance from a flat rigid plate to any portion of the convex surface of a card shall not be greater than 1.5 mm (0.06 in) including the card thickness.
- Card physical characteristics: Bending stiffness, Flammability, Toxicity, Resistance to chemicals, Card dimensional stability and curve with temperature and humidity...should also be considered to meet ISO 7810 Standard.

ISO 7810 Standard combines with other standards to provide standards for interchange cards. Two following figures show position of magnetic tracks on the card. For more information refer to ISO 7810 Standard Document in English.



Dimensions in millimetres (inches)

Figure 8. Physical back-site of card structure

Term	Track 1	Track 2	Track 3
A	5,79 (0.228) maximum	8,33 (0.328) minimum 9,09 (0.358) maximum	11,63 (0.458) minimum 12,65 (0.498) maximum
B	8,33 (0.328) minimum 9,09 (0.358) maximum	11,63 (0.458) minimum 12,65 (0.498) maximum	15,19 (0.598) minimum 15,82 (0.623) maximum
C	$744 \pm 1,00$ (0.293 $\pm$ 0.039)	$7,44 \pm 0,50$ (0.293 $\pm$ 0.020)	$7,44 \pm 1,00$ (0.293 $\pm$ 0.039)
D	6,93 (0.252) minimum	6,93 (0.252) minimum	----

Note: All tracks have a minimum width of 2.54 mm (0.100 inches).

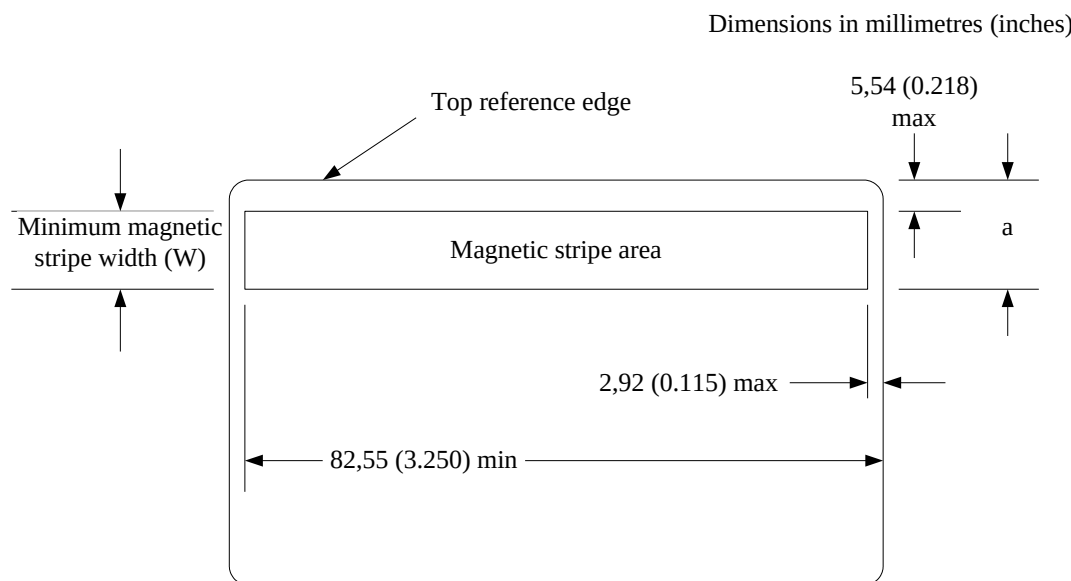


Figure 9. Physical back-site of card structure

- For use of tracks 1 and 2: $a = 11.89 (0.468) \text{ min}$
- For use of tracks 1, 2, and 3: $a = 15.95 (0.628) \text{ min}$

11 ISO 7811

ISO 7811 is a standard that describes the characteristics for identifying cards as well as the necessary elements for joining market of card transactions.

ISO 7811 specifies the information such as: the requirement for the characters to be embossed on the card (ISO 7811-1), magnetic card technology with high impedance and low impedance magnetic card, techniques and encryption technology as well as the set of encoded characters, (ISO 7811-2, ISO 7811-6) ...

ISO 7811-1 described as follows:

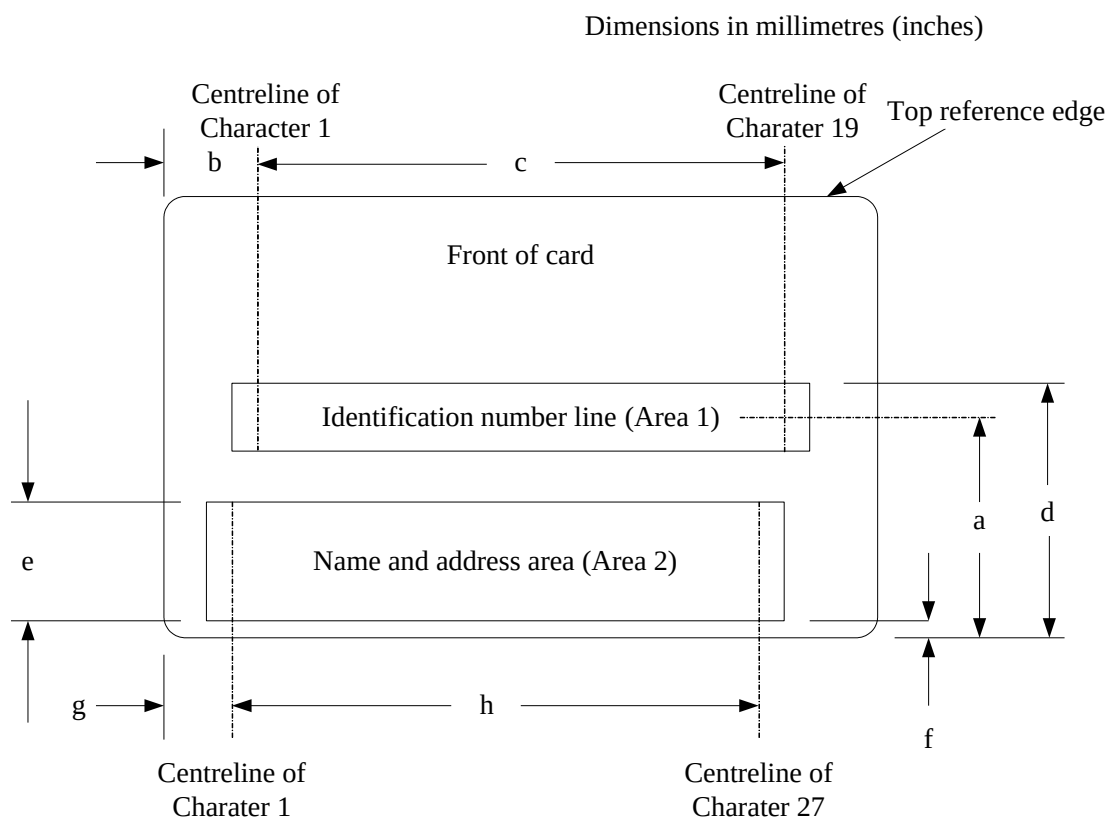


Figure 10. Physical structure front-site of card according to ISO 7811-1

Identification number line (Area 1)		Name and address area (Area 2)	
A	$21,42 \pm 0,12$ (0.843 ± 0.005)	e	14,53 (0.572) maximum
B	$10,18 \pm 0,25$ (0.401 ± 0.010)	f	2,54 (0.100) minimum 3,30 (0.130) maximum
C	$65,31 \pm 0,76$ (2.571 ± 0.030)	g	$7,65 \pm 0,25$ (0.301 ± 0.010)
D	24,03 (0.946) maximum	h	$66,04 \pm 0,76$ (2.600 ± 0.030)

There are 02 embossed area on card:

- Area 1 is used to emboss identification number of card
- Area 2 is used to emboss name, address related to cardholder

Area 1 is embossed on single line, maximum is 19 characters with follow feature:

- Uses one in 03 sets of character for embossing:

ISO 1073 -1, OCR-A, Sizes I and IV.

ISO 1073 -2, OCR-B, Sizes I and IV.

Farrington 7B (character sets commonly used to emboss)

- The distance from the center of one character to another character is 0.15 ± 3.63 mm is $(0.143 \pm 0.006$ in).
- The maximum height of a character embossed is 4.32 mm (0.170 in).
- The thickness of the embossed characters is from the unembossed position to the highest position of embossed characters. The following table gives information about the thickness of embossed characters of unused cards (cards with full information but not issued for use) and returned card (the card has been used by the cardholder but is returned for the purpose of testing).

	Card Identification Number (Area 1)		Cardholder Information (Area 2)	
	Maximum	Minimum	Maximum	Minimum
Unused card	0,48 (0.019)	0,40 (0.016)	0,46 (0.018)	0,36 (0.014)
Returned card	0,48 (0.019)	0,30 (0.012)	0,46 (0.018)	0,26 (0.010)

The information on the card holder (Area 2), such as name, address ... will be embossed on 4 lines with a maximum of 27 characters. Character set "ISO 1073-2, OCR-B, I and IV Size" will be used to emboss. These properties should be noted for the area that contains information about the name and address of the cardholder:

- The distance from the center of one character to another character is 2.54 ± 0.15 mm (0.100 ± 0.006 in).
- The thickness of the embossed characters is from the unembossed position to the highest position of embossed characters. The following table gives information about the thickness of embossed characters of unused cards and returned cards.

Details about the character set used for embossing and other characteristics information, refer the appendix in the English version of ISO 7811-1.

12 ISO 7811-2 and ISO 7811-6

These standards correspond the low coercivity and high coercivity magnetic stripe cards, the difference between the two standard in the material standard material made from card and card durability. For properties such as physical location, density of bits, the encoded characters, the two standards are similar.

These standards stipulate card has 03 tracks with:

- **Track 1** is the track that complies with IATA (International Air Ban sport Association). This is a read-only track, recorded at high density and can contain both numbers and alphabetic characters.
- **Track 2** is the track that complies with ABA track (America Banker Association). This is a read-only track, recorded at low density and contains only numeric characters.
- **Track 3** is the track that complies with TTS (Thift Third) with high recording density, contains only numeric characters but it is able to overwrite data on the components has been existed.

Information on the property and recording density on each Track of card... can be summarized as follows:

Track	Property	Density	Presented	Length	Code format	Number of characters
Track 1	Read only	210 bits/inch	Character and number	79 characters	7 bits (6 data bit + 1 bit checks even and odd)	$2^6=64$
Track 2	Read only	75 bits/inch	Number (0→9)	40 characters	5 bits (4 data bit + 1 bit checks even and odd)	$2^4=16$
Track 3	Read, write	210 bits/inch	Number (0→9)	107 characters	5 bits (4 data bit + 1 bit checks even and odd)	$2^4=16$

Two ISOs outlined in this section, there are many other features such as the characters are defined on the tracks, the value of the initial / end information... these properties were presented very clearly in the "Information stored on the Tracks"; therefore within the scope of the document will not be mentioned in this appendix. The above information is extracted from the English on the ISO 7811-2 and ISO 7811-6.

13 ISO 7812

ISO 7812 is a set of standards defines on the numbering system on the card and the necessary procedures for signing up to become a card issuer. PAN (Primary Account Number) is the unique identification for each card. PAN can be called with other names such as card numbers or account numbers.

Card number or account number of the card can be up to 19 numbers, but here would introduce 16-digit is made up of three components as follows:

IIN fixed length 6 digits assigned by NAPAS Registration Authority						<u>Customer Account Number</u> assigned by card issuer (Variable up to 12 digits including any spaces) Digits 7 – 15										Check Digit
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	

- IIN - Issuer Identification Number: This is the identification number of issuer known as the BIN (Bank Identification Number). It includes 06-digit number. NAPAS will be based on the BIN to identify issuing bank. In this six digits may include the following components:
- MII - Major Industry Identifier: This is the number used to specify in the field service industry. This number has value which complies as follows:
 - 0: To reserve for future use by ISO / TC 68.
 - 1: For aviation organizations.
 - 2: Assigned to the organization for aviation in the present and future.
 - 3: For the organization of tourism and recreation.
 - 4: For financial institutions and banks.
 - 5: For financial institutions and banks.
 - 6: For the banking sector as well as in the areas of trade.
 - 7: For the organization related to petroleum.
 - 8: For those organizations related to the field of communications.
 - 9: To reserve for domestic.
- II - Issuer Identifier: The identification of the card issuer, this can have up to 05 digits so. In this context, the first number (MI) in the digits of IIN shall be prescribed by the subsequent components, for example if MII = 9, the next 03 numbers should contain the country code (CC - country code).

- IAI - Individual Account Identification: Number used to identify client accounts, because here the PAN is 16 digits therefore it should have 9 digits. Banks may provide information structures in this field.
- CD - Check Digit: This is used to check the number that exists in front of it, this includes a digit and this is generated from applying of the Luhn algorithm.

Hereafter is typical structure for the 16-digit card number:

- The card number includes 16 digits.
- The card is unique to each customer.
- The bank will register a separate bank formats (Bank Identification Number - BIN). Via BIN, switching system knows where messages routed to.
- When used, the bank will use this BIN number.
- The branch will register branch digit, it includes 3 digits. This three digits will be part of the card. When looking at the card, it can identify it belong to which branch.
- PAN's structure as follows:

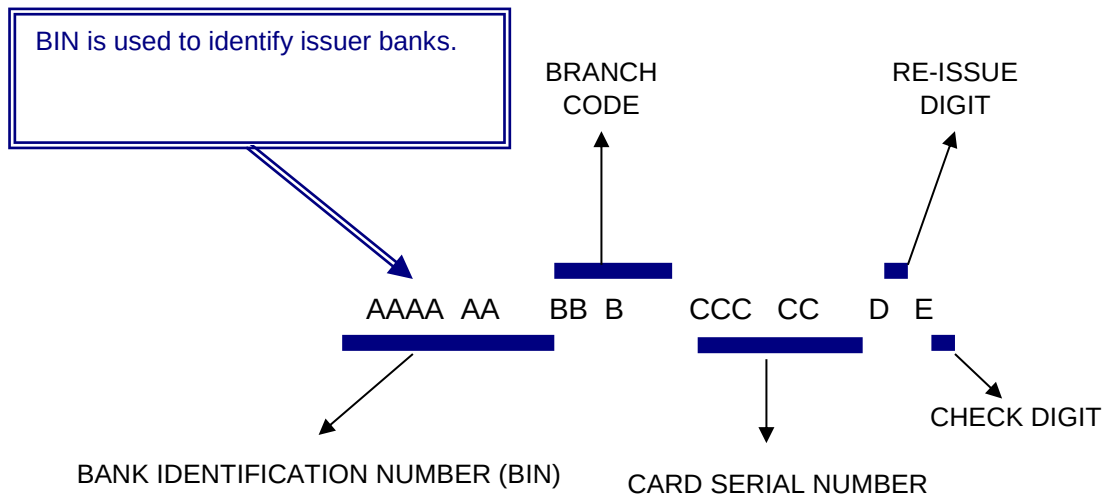


Figure 11. Figure 15. Typical structure of card number

Number	Length	Name	Description
AAAA AA	6 (six)	Bank Identification Number (BIN)	Register number of bank provided by ISO Only for banks
BB B	3 (three)	Branch code	To specify cardholder of which branch Bank may have maximum 999 branches
CCC CC	5 (five)	Card number	Specifying registration of card Each customer has private number Each branch may have up to 99999 customers Serial number registration is started from 1 and it increases 1 character. Registration number calculated by each branch. (each branch has own serial number)
D	1 (one)	Number of re-issuing	To show number of issuing The first issuing, its value is 0 Each time of re-issuing, it increase 1 unit.

E	1 (One)	Checking number	Checking number, it is used to security purpose.
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For more information, please refer to English version of ISO 7812

14 ISO 7813

ISO 7813 is a set of standards which stipulates about the structure of the bar (the tracks of the card) and the content of information stored on the bar. Hereafter is introduction of the track 1 and track 2. It provides information to exchange between members who take part in the switching network system:

14.1 Track 1

Card Data Format - Track 1

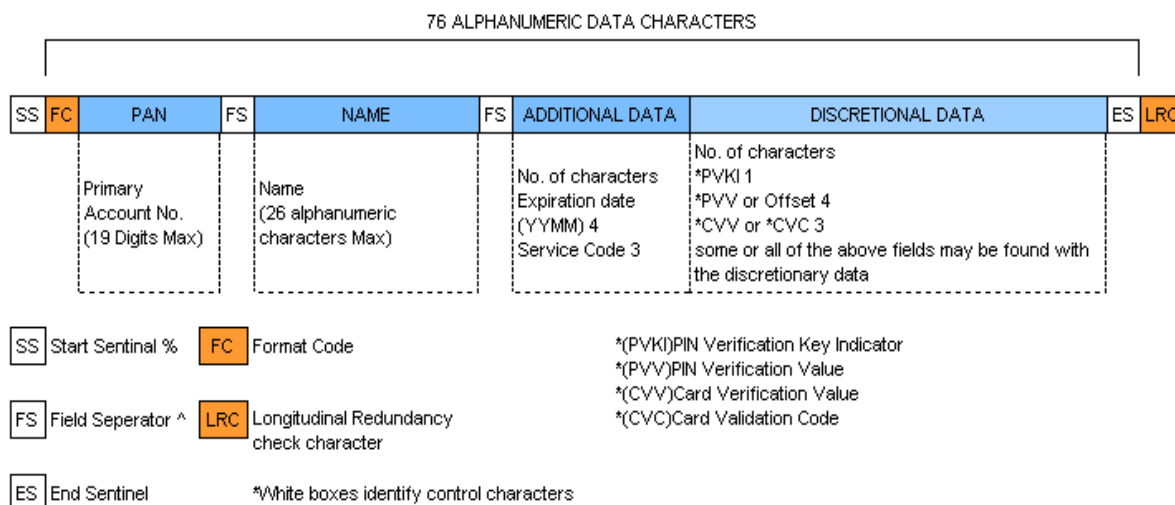


Figure 12. Figure 16. Data structure of Track 1 compliant ISO 7813

ISO 7811 and ISO 7813 stipulate information for track 1 as below:

Density recording on the card: 210 BPI (bits per inch).

Character format: 7 bits per character (including 6 bits of data and 1 parity check bit).

Information about the content: can contains a maximum of 79 characters (both word and numeric (alphanumeric)).

From the data format for track 1, it shows, 79 characters in track 1 format includes many fields such as SS, FC, FS, ES ... With the track 1, the information is only readable. Following are details of the field in track 1 are listed in sequential that track.

SS - Start Sentinel (characters mark the start of track): contains one character, from the data format table for ANSI / ISO Alpha, it has a character is, "%". (Refer to characters table of track 1)

FC - Format Code (Identifier): is a character (but only the alphabetic characters, cannot be a number or special character), the meaning of this character as follows:

- A: to be used for private purposes by the card issuer.
- B: to be used in the field of banking and finance practice. This is the code that the banks participate in the switching network generally follow.
- C-M: for use by ANSI Subcommittee X3B10.
- N-Z: available for individual card issuers.

PAN - Primary Account Number: This field complies with ISO 7812 and is introduced in the previous section.

FS - Field Separator: This is the character to separate between the fields, according to ISO, **for track 1, this character is "^"**.

CC - Country Code: country code, it includes 3 digits and is introduced at ISO 3166. This field has value is 704 to express the country code of Vietnam.

NM – Name: this field may have from 2-26 characters, including:

- *Surname(s): is separated by a blank space (if necessary).*
- Surname separator: /.
- Suffix.
- First name(s).
- Title separator (.) (*When title field is presented and encoded*)
- *Title is option field.*

If this field is not used, a space character and characters "/" followed will be filled in.

FS - Field Separator: This is a character used to split between fields. In track 1, character separates between the characters is "^".

ED - Expiry Date: This is field used to specify expiration date of the card, it includes 4 digits with format: YYMM. If this field is not used, then a separating character will be replaced on it.

SC - Service Code: The service code has 3 digits that defines the different types of services, different types of cards are used in domestic and foreign exchanges, the requirements specified for PIN ... The digits in this field have the following meanings:

The first character (it has the most meaning): with transaction exchange_purpose

- 0: reserved for future use by ISO.
- 1: For the exchange of international use.
- 2: For the exchange of the international use of the integrated circuit on the card, which means it is used for financial transactions for smart cards.
- 3: It is intended for future use for the ISO.
- 4: It is intended for future use for the ISO.
- 5: Available for exchange transactions in domestic, excepting it has been agreed bilaterally between the two countries.
- 6: Available for exchange transactions in domestic, excepting it has been agreed bilaterally between the two countries, is assigned to the case with an integrated circuit used in the financial transactions.
- 7: For private cards, not for the exchange of common use, exception agreed bilaterally.
- 8: Reserved for future use ISO.
- 9: For testing.

The Second digit: It has meaning for authorization process:

- 0: Authorized transactions with features of common law.
- 1: Reserved to use for future ISO.
- 2: The transaction is authorized online by the card issuer.
- 3: Reserved to use for future ISO.
- 4: The transaction is authorized online by the card issuer, except bilateral agreement between two parties.
- 5: Reserved to use for future ISO.

- 6: Reserved to use for future ISO.
- 7: Reserved to use for future ISO.
- 8: Reserved to use for future ISO.
- 9: Reserved to use for future ISO.

The third digit: (it has the least meaning): The range of services and requirements about PIN:

- 0: Do not limit the scope of the service but the PIN is required.
- 1: Do not limit the scope of the service.
- 2: Only goods and services (non-cash).
- 3: Only for transactions on ATMs and PIN is required.
- 4: Only for cash transactions.
- 5: Only for goods and services (non-cash) and PIN is required.
- 6: Unlimited service and PIN is required if possible.
- 7: Only for goods and services (non-cash) and PIN is required if possible.
- 8: Reserved to use for future ISO.
- 9: Reserved to use for future ISO.

If this field is not used, then an FS separate character will be replaced in its position.

PVV - PIN Verification Value: PVV includes 5-digits, this is a feature brings additional information to ensure more safety.

DD - Discretionary Data: The discretionary data, this is the rest of the message, the magnitude of this field depends on the elasticity of all previous fields to ensure track 1 always contains 79 characters (for example, if as the PVV does not exist, the number of characters for this field including the number of characters PVV). This field has different characteristics for different organizations, or it may have private features of the card issuers. This field may contain information on which card issuers use their on-us transactions. On his discretionary data, the card issuer can add a lot of sub-fields contains different information, serving for many different purposes

ES - End Sentinel: Character marks the end of track 1, that character is denoted by "?".

LRC - LongituBM #Redundancy Check: This is a character used to check all the data in the track 1. This character was generated depending on the algorithm applied to the test.

14.2 Track 2

Card Data Format - Track 2

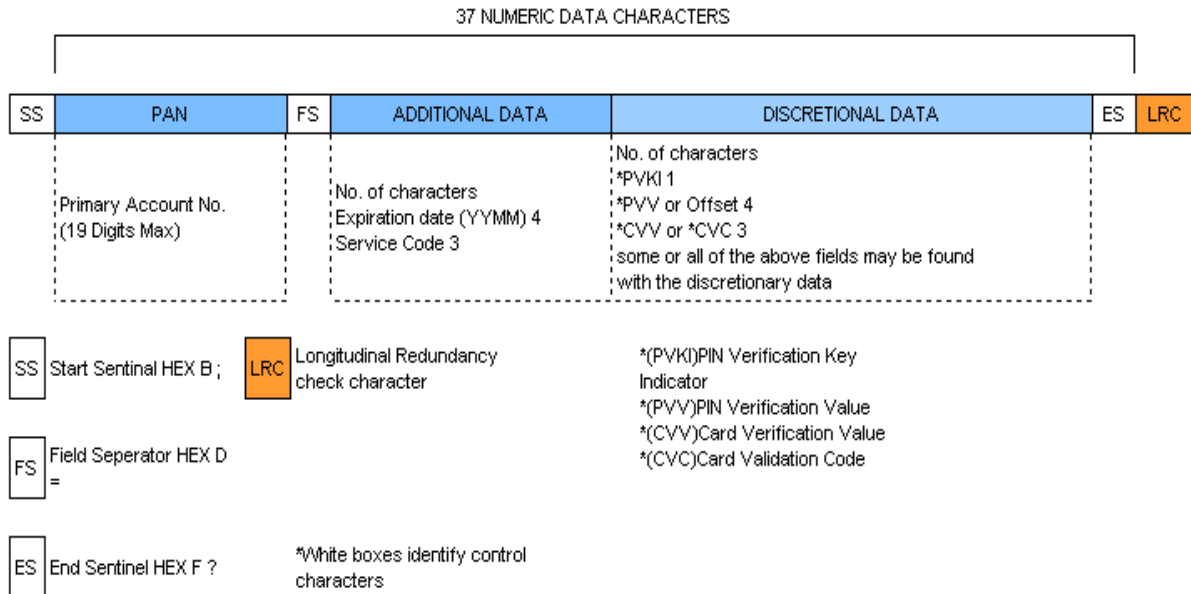


Figure 13. Figure 18. The data structure of Track 2 according to ISO 7813

ISO 7811 and ISO 7813 specifies the information to track 2 as follows:

Recording density on the card: 210 BPI (bits per inch).

Character format: 5 bits per character (including 4 bits of data and 1 parity check bit).

Information about the content: It may contain up to 40 characters (just numeric).

From the data format for track 2 shows that, 40 characters in track 1 format includes multiple fields such as SS, FC, FS, ES ... With track 2, the information is only readable. Following are details of field of the track 2 is listed sequentially.

SS - Start Sentinel (character marks the start of track): contains a character, from the data format table for ANSI / ISO Alpha it show that it is character is ";".

PAN - Primary Account Number: This field complies with ISO 7812 introduced in the previous section.

FS - Field Separator: This is the separating character between the fields, according to ISO, this character is "=".

CC - Country Code: country code, it includes 3-digits and is mentioned in ISO 3166. The value of this field is equal 704 is Vietnamese code.

ED - Expiry Date: This is the specified expiration date of the card, it includes 4-digit format: YYMM. If this field is not used, then a separator character will be replaced on it.

SC - Service Code: This field is called the service code. Detail information on this field, please see the section of Track 1.

PVV - PIN Verification Value: It includes 5-digits, this is a feature brings additional information to ensure more safety.

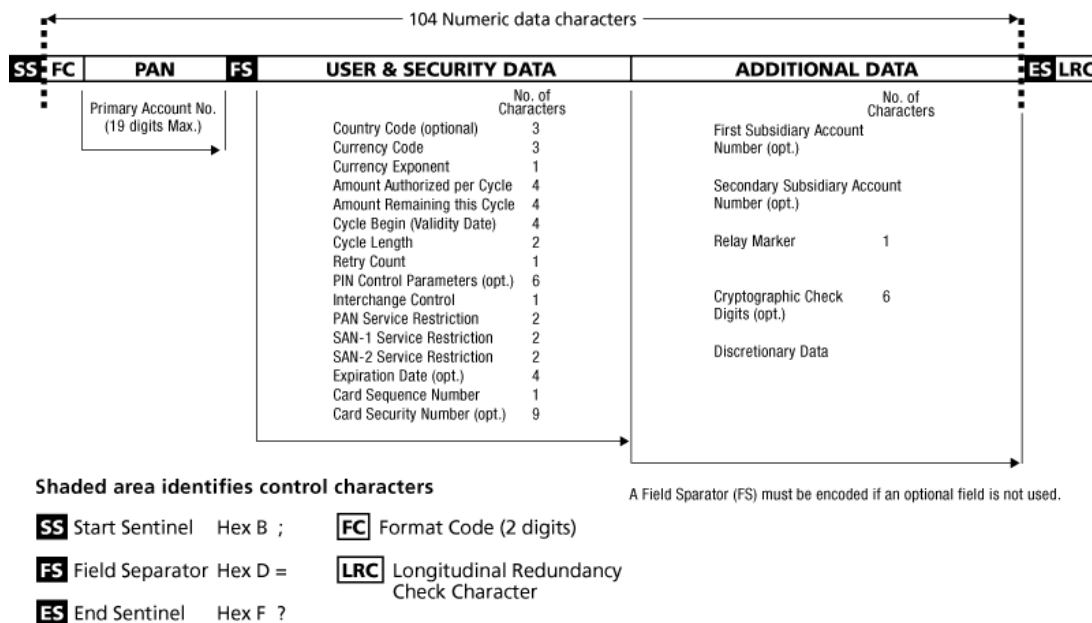
DD - Discretionary Data: The discretionary data, this is the rest of the message, the magnitude of this field depends on the elasticity of all previous fields to ensure track 1 always contains 79 characters (for example, if as the PVV does not exist, the number of characters for this field including the number of characters PVV). This field has different characteristics for different organizations, or it may have private features of the card issuers. This field may contain information on which card issuers use their on-us transactions. On his discretionary data, the card issuer can add a lot of sub-fields contains different information, serving for many different purposes.

ES - End Sentinel: Character marks the end of track 2, this denotes as "?".

LRC - Longitudinal Redundancy Check: This is one character used to check all the data in the track 2. This character was generated depending on the algorithm applied to the test.

For more information please refer to the English version of ISO 7813.

15 ISO 4909



ISO 4909 defines information for track 3 as below:

Recording density on card: 210 dpi.

Character format: 5 bits per character (including 4 bits of data and 1 parity check bit).

Information about the content: It may contain up to 107 characters (just numeric).

From the data format for track 3 shows that, 107 characters in track 3 format includes multiple fields such as SS, FC, FS, ES ... With track 3, the information is readable and writable. Following are details of field of the track 3 is listed sequentially.

SS - Start Sentinel (character marks the start of track): contains a character, from the data format table for ANSI / ISO Alpha it show that it is character is ";".

FC – Format Code: including 2 digits

- 00: Not valid for international interchange.
- 01-02: Bank/financial. These are the formats described here.
- 03-19: Reserved for future use by ISO/TC 68.
- 20-89: Reserved for future use by ISO/TC 95 SC 17.

- 90-99: Reserved for proprietary use of card issuer, but not for international interchange.

PAN - Primary Account Number: This field complies with ISO 7812 was introduced in the above section. If the track 3 is used with the track 2, the PAN is option information field.

FS - Field Separator: This is the separating character between the fields, according to ISO, this character is "=".

CC - Country Code: country code, it includes 3-digits and is presented at ISO 3166. The value of this field is equal 704 that define a Vietnamese code. If this field is not used, the FS will be filled.

CuC - Currency Code: it includes 3-digits. The currency code is introduced in ISO 4217. For example, '704' represents Vietnamese dong, '840' represents US dollar. If 3 digits 0 (000) to be filled in this field means that the card is not valid for international payments.

CE - Currency Exponent: This code consists of one digit.

0-5: power of ten by which multiply the currency amount fields (AA and AR) to get the actual value in the currency of **CuC** field.

AA - Amount Authorized per cycle, it includes 4-digits.

Maximum amount permitted in one cycle.

If there are 4 digits of 0 (0000) recorded in this field means that there is not valid for charge operations (not debit).

AR - Amount Remained of this cycle, it includes 4-digits.

The maximum amount of money permitted in the transaction. This is dynamic field and is initialized with value AA for the first time the card is used in a new cycle. Then the field will be modified accordingly.

CB - Cycle Begin (invalid date): it includes 4-digits.

The date that the transaction cycle actually began. YDDD format in which Y (Year) is the value of the last number of years, DDD (is the day of the year, from 001 to 366). This field must be updated each time a new transaction begins.

In addition, this field can indicate the date from which the card is valid.

CL – Cycle Length; it includes 2 digits.

- 00: Infinite, AR should be decremented but never reset
- 01-79: Number of days
- 80: Cycle begin each 7 days
- 81: Cycle begin each 14 days
- 82: Cycle begins each 1st and 15th days of every month
- 83: Cycle begins the day of the month specified in CB of every month
- 84: Cycle begins the day of the month specified in CB of every third month
- 85: Cycle begins the day of the month specified in CB of every sixth month
- 86: Cycle begins the day of the year specified in CB of every year
- 87-89: Reserved for future use by ISO/TC 68
- 90-99: Reserved for proprietary use of card issuer, but not for international interchange

This field represents the duration of a cycle for which the AA limit holds.

RC - Retry Count: This code recorded 1 digit.

Number of remaining PIN trials, the value is initialized to 3 times and reduced by one unit after every wrong PIN entered. This field is reset to 3 after entering PIN successfully. When this field reaches 0 the card is invalid for any interchange purpose.

PINCP - PIN Control Parameters (PINPARM): This code recorded 6 digits:

If FC = 01 the two first digits represent the algorithm used to calculate PIN, where 00-09 mean private algorithm, 10-19 mean DEA and values 20 to 99 are reserved for future use by ISO/TC 68. Next 4 digits are PIN offset, a complementary value of PIN so customers can change their PIN, or PVV.

If FC = 02 the first digit represents the algorithm used to calculate PIN, where 0 means private algorithm, 1 means DEA and values 2 to 9 are reserved for future use by ISO/TC 68. The second digit represents a key for the algorithm. Next 4 digits are PIN offset, a complementary value of PIN so customers can change their PIN, or PVV.

If this field is not used then FS will be in place.

IC – Interchange Control: this code records one digit.

- 0: No restriction
- 1: Not available for international interchange
- 2-8: Limited interchange, only local use and under agreement
- 9: Limited interchange, recommended for test cards

PANSR – PAN Service Restriction: this code records two digits.

The first character defines the type of account:

- 0: Associated account number not encoded on track.
- 1: Savings account
- 2: Current or checking account
- 3: Credit card account
- 4: Generic or universal account
- 5: Interest-bearing current or checking account
- 6-8: Reserved for future use by ISO/TC 68
- 9: Reserved for card issuer's internal use, not for interchange

The second digit defines the scope of service:

- 0: No restrictions.
- 1: No cash dispense.
- 2: No point of sale (POS) transaction.
- 3: No cash dispense and no POS transaction.
- 4: Authorization required.
- 5-7: Reserved for future use by ISO/TC 68.
- 8-9: Reserved for card issuer's internal use, only local use and under agreement.

FSANSR - FSAN Service Restriction: This code records 2 digits:

The same value and meaning with PANSR fields.

SSANSR: SSAN Service Restriction: This code records 2 digits:

The same value and meaning with PANSR fields.

ED – Expiry Date: this code records 4 digits: YYMM

If this field not be used, FS will be in place.

CSN - Card Sequence Number: this code records one digit:

Allows to distinguish among different cards with the same PAN. It is incremented by one unit in any additional card.

If FC = 02 and FS is filled, it means that the CSN field has the value.

CScN: Card Security Number: this code includes 9 digits:

The first digit represents the algorithm used to calculate the verification value on magnetic track against the embossed characters, where 0-4 are available for national use, 5-8 are available for international security methods given by ISO/TC 68, and 9 is reserved for private use. Next 8 digits are the verification value.

If this field is not used, a FS will be in place.

FSAN: First Subsidiary Account Number. The digit will be changed.

This is optional field.

FS – Field Separator: this code includes 1 digit is “=”.

SSAN - Second Subsidiary Account Number: Variable number of digits

This is optional field.

FS – Field Separator: this code includes 1 digit is “=”.

RM – Relay Marker: this code includes 1 digit.

- 0: Include AD and DD fields in transactions messages.
- 1: Do not include AD field in transactions messages.
- 2: Do not include DD field in transactions messages.
- 3-9: Invalid.

CCD - Crypto Check Digits: this code includes 6 digits.

A validation value which permits integrity verification of the magnetic stripe content.

If this field is not used, FS will be in place.

AD – Additional Data: this code includes a rest of characters.

The optional field used for proprietary of issuer if FC = 01.

If FC = 02, the following sub-field will be required:

TD: Transaction Date: this code includes 4 digits

Date of last cash dispense. The format is YDDD where Y stands for the least significant digit of the year and DDD is the day of the year (001 to 366).

If this field is not used a FS will be in place.

AVV: Additional Verification Value(s): this code includes 8 digits:

Validation of the PIN or two additional PINs corresponding to different keys of the same algorithm. It can be an eight digit value, two four digit numbers or, in connection with the last four digits of the PINCP field, two six digit numbers.

If this field is not used a FS will be in place.

ACSN: Alternative Card Sequence Number: this code includes 3 digits.

This field has the same purpose with CSN field but allow the maximum number is 1000 cards.

If this field is not used, FS will be in place.

INIC: International Network Identification Code: This code includes 3 digits.

Code for identification of an international group of issuers, when the IIN cannot be used. It is equivalent to the term Network International Identifier used in [ISO 8583](#).

If this field is not used, the FS will be in place.

DD - Discretionary Data: this code includes a rest of characters.

Optional field reserved for proprietary use of card issuer.

ES – End Sentinel: including one character is “?”.

LRC - Longitude Redundancy Check: includes one character.

16 ISO 3166

ISO 3166 is ISO code regulation of each country in the world. Country code includes 03 digits from 00 to 09 combined. Vietnam's national code number is "704". Within the scope of this document does not present in detail for each country code of the world. Detailed information about these countries, refer to ISO 3166 standard document in English.

17 ISO 9564

ISO 9564-1 introduces the principles and requirements for processing online PIN for ATMs and POS:

- Describing the principles and basic techniques to provide a minimum level of security to managing PIN management effectively. This standard is compatible with responsible organizations implementing management techniques and protecting PIN.
- Describing the technical to protect PIN appropriately transactions from online environment, and a standard for data exchange of PIN. These techniques are provided for the organization responsible for implementing the technical management and protecting PIN at ATM and POS.

Detail information about these countries, see the ISO 9564 – 1 standard document in English.

18 ISO 13491

ISO 13491 is the ISO defines the concept, the requirements for safe encrypted device used in the system using magnetic cards.

19 ISO 11568

ISO 11568 is the ISO defines the concepts, technologies as well as technical requirements for key management processes.

Detail information about these countries, refer the ISO [ISO 11568 - 1](#), [ISO 11568 - 2](#), [ISO 11568 - 3](#), [ISO 11568 – 4](#), [ISO 11568 – 5](#) standard document in English.

20 Effective document

The NAPAS Technical Specification for switching service takes effect from 01st January, 2017.

21 Document management

Internal reference document:

No.	Document Name
1	BANKNETVN Technical Specification Document (version 1.6.1).
2	SMARTLINK Technical Specification Document (version 2.0).
3	Regulation for Member Organization Participating on the SMARTLINK System.
4	Regulation for Member Organization Participating on the BANKNETVN System.
5	Regulation for Business Settlement and Reconciliation Process on the BANKNETVN System (released at 28th February, 2013).

External reference document:

No.	Document Name
1	ISO 8583-1987 Technical Specification Document.
2	Technical Specification Documents of VISA Organization.
3	Technical Specification Documents of MASTERCARD Organization.
4	Technical Specification Documents of UPI Organization.